

Property A and Operator-Norm Localization: The Complete p -Phase Diagram

Candidate full answer to arXiv:1809.00532, Question 6.6

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Status: candidate full solution, likely valid. For every $1 < p < \infty$, Proposition 6.3 remains true with $\ell^2(X)$ replaced by $\ell^p(X)$. At $p = 1$, its localization clauses are universal and do not characterize Property A.

1 Source question and definitions

Špakula and Zhang prove a commutant version of operator norm localization that characterizes Property A on $\ell^2(X)$, and ask on PDF page 20 of [1]:

Question 6.6. Is it possible to generalise the characterisation of Property A (Proposition 6.3) to any $p \in [1, \infty)$?

Let X be a bounded-geometry metric space and $1 \leq p < \infty$. For $L, \eta > 0$, set

$$\text{Commut}_p(L, \eta) = \{b \in \mathcal{B}(\ell^p(X)) : \|[b, f]\| < \eta \text{ for every } L\text{-Lipschitz } f \in C_b(X), \|f\|_\infty \leq 1\}.$$

The direct p -versions of the three operator clauses in Proposition 6.3 are:

- (L₁^p) For every $\varepsilon, L, M > 0$ there is $s > 0$ such that, if $\|b\| \leq M$ and $b \in \text{Commut}_p(L, \varepsilon/6)$, some unit vector v with $\text{diam}(\text{supp } v) \leq s$ satisfies $\|bv\|_p \geq \|b\| - \varepsilon$.
- (L₂^p) For every $c \in (0, 1)$ and $L > 0$ there is $s > 0$ such that, if $\|b\| = 1$ and $b \in \text{Commut}_p(L, (1 - c)/6)$, some unit vector v with $\text{diam}(\text{supp } v) \leq s$ satisfies $\|bv\|_p \geq c$.
- (L₃^p) (p -ONL) For every $c \in (0, 1)$ and $R > 0$ there is $s > 0$ such that every norm-one operator $a \in \mathcal{B}(\ell^p(X))$ of propagation at most R has a unit vector v with $\text{diam}(\text{supp } v) \leq s$ and $\|av\|_p \geq c$.

For $E \subseteq X$, write $\partial_R E = \{x \in X \setminus E : d(x, E) \leq R\}$. Following [2], X has *uniform local amenability* (ULA) if for every $R, \varepsilon > 0$ there is $S > 0$ such that, for every finite $F \subseteq X$, some $E \subseteq X$ with $\text{diam}(E) \leq S$ satisfies

$$|\partial_R E \cap F| < \varepsilon |E \cap F|. \tag{ULA}$$

2 Idea of the proof

The source already proves Property A $\Rightarrow (L_1^p)$ for every $1 < p < \infty$. The missing direction is geometric. Suppose ULA fails. At one fixed scale R , arbitrarily large localization radii then admit a finite set F in which every small-diameter subset has a uniformly large outer vertex boundary. Join points of F at distance at most R and consider the random walk P on this finite bounded-degree graph.

The lazy walk $(I + P)/2$ has norm one on every weighted ℓ^p because it fixes constants. On the other hand, a coarea estimate turns local vertex expansion into a local ℓ^2 Cheeger inequality. The scalar monotonicity inequality for $z \mapsto |z|^{p-2}z$ then gives a uniform lower bound for $\|(I - P)v\|_p$ on every small-support vector. Uniform convexity of ℓ^p forces $(I + P)/2$ to lose a fixed amount of norm on all such vectors. Conjugating from degree measure to counting measure produces a finite-propagation operator that violates p -ONL. Thus p -ONL implies ULA, and Elek's theorem [3] then gives Property A.

At $p = 1$, uniform convexity disappears for a decisive reason: every operator is almost normed by a single column. Hence all three localization clauses are automatic.

3 The local lazy-walk estimate

For a finite graph $G = (F, \mathcal{E})$ with no isolated vertices, write d_x for the degree of x and

$$\|w\|_{p,d}^p = \sum_{x \in F} d_x |w_x|^p, \quad (Pw)_x = \frac{1}{d_x} \sum_{y \sim x} w_y.$$

Jensen's inequality shows that P is a contraction on $\ell^p(F, d)$.

Lemma 1 (Local Cheeger estimate). *Suppose G has maximum degree at most D , and $K \subseteq F$ has the property that*

$$|\partial_G A| \geq h|A| \tag{1}$$

for every nonempty $A \subseteq K$, where $\partial_G A$ is the outer vertex boundary in F . If $g \geq 0$ is supported in K , then

$$\sum_{\{x,y\} \in \mathcal{E}} |g_x - g_y|^2 \geq \frac{h^2}{2D^2} \sum_{x \in F} d_x g_x^2. \tag{2}$$

Proof. For $A_t = \{x : g_x > t\}$, (1) implies that the number of edges leaving A_t is at least $h|A_t|$. The layer-cake formula therefore gives

$$\sum_{\{x,y\} \in \mathcal{E}} |g_x^2 - g_y^2| = \int_0^\infty 2t |\mathcal{E}(A_t, A_t^c)| dt \geq h \sum_x g_x^2. \tag{3}$$

By Cauchy–Schwarz and the degree bound, the left side of (3) is at most

$$\left(\sum_{\{x,y\} \in \mathcal{E}} |g_x - g_y|^2 \right)^{1/2} \left(2D \sum_x g_x^2 \right)^{1/2}.$$

Squaring and using $\sum_x d_x g_x^2 \leq D \sum_x g_x^2$ proves (2). $\square$

Lemma 2 (Uniform local loss for the lazy walk). *Fix $1 < p < \infty$, $D \geq 1$, and $h > 0$. There is $q = q(p, D, h) < 1$ with the following property. Under the hypotheses of Lemma 1, every $w \in \ell^p(F, d)$ supported in K satisfies*

$$\left\| \frac{I + P}{2} w \right\|_{p,d} \leq q \|w\|_{p,d}. \quad (4)$$

Proof. For $z, u \in \mathbb{C}$ and $1 < p < \infty$, monotonicity of the duality map gives

$$\operatorname{Re}[(z - u)(|z|^{p-2}\bar{z} - |u|^{p-2}\bar{u})] \geq \gamma_p(|z|^{p/2} - |u|^{p/2})^2, \quad \gamma_p = \frac{4(p-1)}{p^2}. \quad (5)$$

For complex scalars, the left side is minimized when z and u have the same argument. For $a \geq b \geq 0$, Cauchy–Schwarz gives

$$\left(\int_b^a t^{p/2-1} dt \right)^2 \leq (a - b) \int_b^a t^{p-2} dt,$$

which, after evaluating the two integrals, is exactly the resulting real inequality in (5).

Pair $L = I - P$ against the $\ell^p(F, d)$ duality map. Symmetrizing over unoriented edges, applying (5), and then Lemma 1 to $g_x = |w_x|^{p/2}$ gives

$$\begin{aligned} \|Lw\|_{p,d} \|w\|_{p,d}^{p-1} &\geq \operatorname{Re} \sum_x d_x (Lw)_x |w_x|^{p-2} \bar{w}_x \\ &\geq \gamma_p \sum_{\{x,y\} \in \mathcal{E}} (|w_x|^{p/2} - |w_y|^{p/2})^2 \\ &\geq \frac{\gamma_p h^2}{2D^2} \|w\|_{p,d}^p. \end{aligned} \quad (6)$$

Thus $\|w - Pw\|_{p,d} \geq \delta \|w\|_{p,d}$, where $\delta = \gamma_p h^2 / (2D^2) > 0$. Since P is a contraction and $\ell^p(F, d)$ has the same positive modulus of uniform convexity as an ℓ^p space, (6) implies

$$\left\| \frac{w + Pw}{2} \right\|_{p,d} \leq (1 - \rho_p(\delta)) \|w\|_{p,d},$$

where $\rho_p(\delta) > 0$ depends only on p and δ . This is (4). $\square$

4 The new reverse implication

Theorem 3. *Let X have bounded geometry and let $1 < p < \infty$. If X has p -ONL, then X has ULA.*

Proof. Suppose ULA fails. Then there are $R > 0$ and $h > 0$ such that for every $S > 0$ there is a finite $F \subseteq X$ satisfying

$$|\partial_R E \cap F| \geq h |E \cap F| \quad (7)$$

for every $E \subseteq X$ with $\operatorname{diam}(E) \leq S$. Decreasing h if necessary, assume $h \leq 1$.

On F , form the graph G_F by joining distinct x, y when $d(x, y) \leq R$. Its maximum degree is at most $D = \sup_x |B(x, R)| - 1$. Taking $E = \{x\}$ in (7) shows that G_F has no isolated vertices. If $A \subseteq F$ has diameter at most S , then $\partial_{G_F} A = \partial_R A \cap F$, so (7) gives (1).

Let P_F be the random walk of G_F and $A_F = (I + P_F)/2$ on $\ell^p(F, d)$. Lemma 2 supplies a number $q < 1$, depending only on p, R, h and the bounded-geometry bound, such that

$$\|A_F w\|_{p,d} \leq q \|w\|_{p,d} \quad (8)$$

whenever $\text{diam}(\text{supp } w) \leq S$.

The isometry $U_F : \ell^p(F) \rightarrow \ell^p(F, d)$ defined by $(U_F v)_x = d_x^{-1/p} v_x$ preserves supports. Define on $\ell^p(X)$

$$T_F = U_F^{-1} A_F U_F \quad \text{on } \ell^p(F), \quad T_F = 0 \quad \text{on } \ell^p(X \setminus F).$$

The operator T_F has propagation at most R and norm one: A_F is a contraction and fixes the constant function. For every v of support diameter at most S , (8) applied to $U_F(v|_F)$ yields

$$\|T_F v\|_p \leq q \|v|_F\|_p \leq q \|v\|_p. \quad (9)$$

Choose $c \in (q, 1)$. For the fixed propagation R , equation (9) gives, for every proposed localization radius S , a norm-one operator T_F for which no S -localized unit vector reaches c . This contradicts p -ONL. $\square$

Corollary 4. *For a bounded-geometry metric space and $1 < p < \infty$, p -ONL implies Property A.*

Proof. By Theorem 3, X has ULA. We record why Elek's graph theorem [3] applies without a hidden coarse-geodesicity assumption. Fix $R > 0$ and let G_R have vertex set X and edges between distinct points at distance at most R . Its degree is bounded by some D . Given $\varepsilon > 0$ and a finite subgraph H on a vertex set F , metric ULA at scale R , with tolerance ε/D , produces E of uniformly bounded diameter with $|\partial_R E \cap F| < (\varepsilon/D)|E \cap F|$. Put $A = E \cap F$. Bounded geometry uniformly bounds $|A|$. Every inner boundary vertex of A in H has a neighbor in $\partial_R E \cap F$, and each such outer vertex is charged at most D times. Hence $|\partial_H^{\text{in}} A| < \varepsilon|A|$, so the class of finite subgraphs of G_R is uniformly locally amenable. Elek's theorem gives Property A for G_R . Its controlled supports in graph distance are controlled in X -distance, and every pair at X -distance at most R is an edge of G_R . Applying this for each R gives Property A for X (the finite case being immediate). $\square$

5 Completion of the characterization

Lemma 5 (Finite propagation implies slow commutation). *For every $R, \eta > 0$ there is $L > 0$ such that every norm-one operator on $\ell^p(X)$ of propagation at most R belongs to $\text{Commut}_p(L, \eta)$.*

Proof. Bounded geometry lets one decompose the controlled relation $\{(x, y) : d(x, y) \leq R\}$ into $N = N(X, R)$ graphs of partial translations. Accordingly, every such operator has a decomposition $a = \sum_{k=1}^N M_{g_k} V_k$, where V_k is a partial translation of propagation at most R and $\|g_k\|_\infty \leq 1$; this is the same support decomposition used in Lemma 6.2 of [1], and it is independent of p . If f is an L -Lipschitz contraction, then

$$\|[M_{g_k} V_k, f]\| \leq LR, \quad \|[a, f]\| \leq NLR.$$

Take $L < \eta/(NR)$ (with the propagation-zero case immediate). $\square$

Theorem 6 (Complete answer to Question 6.6). *Let X be a bounded-geometry metric space.*

- (i) *For every fixed $1 < p < \infty$, the following are equivalent: Property A; the metric sparsification property; and $(\mathcal{L}_1^p)$, $(\mathcal{L}_2^p)$, $(\mathcal{L}_3^p)$ above.*
- (ii) *At $p = 1$, the three localization conditions $(\mathcal{L}_1^1)$, $(\mathcal{L}_2^1)$, $(\mathcal{L}_3^1)$ hold for every metric space, so none of them characterizes Property A.*

Proof. Fix $1 < p < \infty$. Property A is equivalent to metric sparsification, and metric sparsification implies (L_1^p) by the p -version of the commutant localization lemma proved in Section 5 of [1]. The implication $(L_1^p) \Rightarrow (L_2^p)$ follows by taking $M = 1$ and $\varepsilon = 1 - c$. Lemma 5 gives $(L_2^p) \Rightarrow (L_3^p)$. Finally, Corollary 4 gives $(L_3^p) \Rightarrow$ Property A.

For $p = 1$, every $b \in \mathcal{B}(\ell^1(X))$ satisfies the exact column formula

$$\|b\| = \sup_{x \in X} \|b\delta_x\|_1. \quad (10)$$

Indeed, the reverse inequality follows from $\|bv\|_1 \leq \sum_x |v_x| \|b\delta_x\|_1$; the other inequality is immediate. Thus a one-point-supported unit vector ε -norms every b , and it exceeds any fixed $c < 1$ when $\|b\| = 1$. All three localization clauses hold with localization diameter zero, without their commutator or propagation hypotheses. A coarse disjoint union of bounded-degree expanders has bounded geometry but not Property A, proving failure of characterization at $p = 1$. $\square$

6 Novelty, limitations, and verification

Theorem 6 answers the direct formulation of Question 6.6 across its full stated range. It does not claim that no differently designed endpoint condition could characterize Property A at $p = 1$.

Chung and Nowak [4] prove that Property A implies p -ONL for every $p \in [1, \infty)$ and explicitly state that they do not know whether the converse holds, suggesting ULA as a possible route. A bounded search used the exact source question, “ p -operator norm localization”, “independent of p ”, “Property A converse”, “uniform local amenability”, the cited-paper graph, and the run’s cheap indexes. No later resolution through 27 August 2026 was located. Priority is therefore plausible, not certified.

The proof has no computational dependency. The main checks are: the outer boundary in failed ULA is exactly the graph boundary used in (7); all constants in Lemmas 1–2 depend only on p, R, h and bounded geometry, never on F or S ; degree conjugation is an isometry and preserves supports; and the source’s forward commutant lemma already covers every $1 < p < \infty$. Human review should concentrate on these four points and the metric-to-graph application of Elek’s theorem.

Source-page evidence

Question 6.6. Is it possible to generalise the characterisation of Property A (Proposition 6.3) to any $p \in [1, \infty)$?

Second, in the case of $p \in (1, \infty)$ or especially when $p = 2$, is the assumption of Property A really necessary for all the quasi-local operators to be approximable by operators with finite propagation? In other words:

Question 6.7. Is there an example of a space X with bounded geometry and a quasi-local operator on (say) $\mathfrak{B}(\ell^2(X))$ which does not belong to the uniform Roe algebra $B_u^2(X)$ (usually denoted $C_u^*(X)$ in the literature)?

REFERENCES

- [1] G. Bell and A. Dranishnikov. Asymptotic dimension. *Topology Appl.*, 155(12):1265–1296, 2008.
- [2] Jacek Brodzki, Graham A. Niblo, Ján Špakula, Rufus Willett, and Nick Wright. Uniform local amenability. *J. Noncommut. Geom.*, 7(2):583–603, 2013.
- [3] Xiaoman Chen, Romain Tessera, Xianjin Wang, and Guoliang Yu. Metric sparsification and operator norm localization. *Adv. Math.*, 218(5):1496–1511, 2008.
- [4] Yeong Chyuan Chung and Kang Li. Rigidity of ℓ^p Roe-type algebras. *preprint, arXiv:1802.08921*, to appear in *Bulletin of the London Mathematical Society*, 2018.
- [5] Alexander Dranishnikov and Michael Zarichnyi. Asymptotic dimension, decomposition complexity, and Haver’s property C. *Topology Appl.*, 169:99–107, 2014.
- [6] Alexander Engel. Index theory of uniform pseudodifferential operators. *arXiv preprint arXiv:1502.00494*, 2015.
- [7] Alexander Engel. Rough index theory on spaces of polynomial growth and contractibility. *arXiv preprint arXiv:1505.03988*, 2015.
- [8] Vladimir Georgescu. On the structure of the essential spectrum of elliptic operators on metric spaces. *J. Funct. Anal.*, 260(6):1734–1765, 2011.
- [9] Vladimir Georgescu. On the essential spectrum of elliptic differential operators, 2017.
- [10] Vladimir Georgescu and Andrei Iftimovici. Localizations at infinity and essential spectrum of quantum Hamiltonians. I. General theory. *Rev. Math. Phys.*, 18(4):417–483, 2006.
- [11] Mikhael Gromov. Groups of polynomial growth and expanding maps. *Inst. Hautes Études Sci. Publ. Math.*, (53):53–73, 1981.
- [12] Erik Guentner, Romain Tessera, and Guoliang Yu. A notion of geometric complexity and its application to topological rigidity. *Invent. Math.*, 189(2):315–357, 2012.
- [13] Erik Guentner, Romain Tessera, and Guoliang Yu. Discrete groups with finite decomposition complexity. *Groups Geom. Dyn.*, 7(2):377–402, 2013.
- [14] Raffael Hagger, Marko Lindner, and Markus Seidel. Essential pseudospectra and essential norms of band-dominated operators. *J. Math. Anal. Appl.*, 437(1):255–291, 2016.
- [15] Bernhard Hanke, Daniel Pape, and Thomas Schick. Codimension two index obstructions to positive scalar curvature. *Ann. Inst. Fourier (Grenoble)*, 65(6):2681–2710, 2015.
- [16] B. V. Lange and V. S. Rabinovich. Noethericity of multidimensional discrete convolution operators. *Mat. Zametki*, 37(3):407–421, 462, 1985.
- [17] Kang Li, Zhijie Wang, and Jiawen Zhang. A quasi-local characterisation of L^p -Roe algebras. *arXiv preprint arXiv:1808.08593*, 2018.
- [18] Marko Lindner. *Infinite matrices and their finite sections*. Frontiers in Mathematics. Birkhäuser Verlag, Basel, 2006. An introduction to the limit operator method.
- [19] Marko Lindner and Markus Seidel. An affirmative answer to a core issue on limit operators. *J. Funct. Anal.*, 267(3):901–917, 2014.
- [20] N. Christopher Phillips. Crossed products of L^p operator algebras and the K -theory of Cuntz algebras on L^p spaces. *preprint, arXiv:1309.6406*, 2013.
- [21] Vladimir Rabinovich, Steffen Roch, and Bernd Silberman. *Limit operators and their applications in operator theory*, volume 150 of *Operator Theory: Advances and Applications*. Birkhäuser Verlag, Basel, 2004.
- [22] John Roe. An index theorem on open manifolds. I, II. *J. Differential Geom.*, 27(1):87–113, 115–136, 1988.

References

- [1] J. Špakula and J. Zhang, *Quasi-Locality and Property A*, J. Funct. Anal. 278 (2020), 108299; arXiv:1809.00532.
- [2] J. Brodzki, G. A. Niblo, J. Špakula, R. Willett, and N. Wright, *Uniform local amenability*, J. Noncommut. Geom. 7 (2013), 583–603; arXiv:1203.6169.
- [3] G. Elek, *Uniform Local Amenability implies Property A*, Proc. Amer. Math. Soc. 149 (2021), 2573–2577; arXiv:1912.00806.
- [4] Y. C. Chung and P. W. Nowak, *Expanders are counterexamples to the ℓ^p coarse Baum–Connes conjecture*, J. Noncommut. Geom. 17 (2023), 305–331; arXiv:1811.10457.